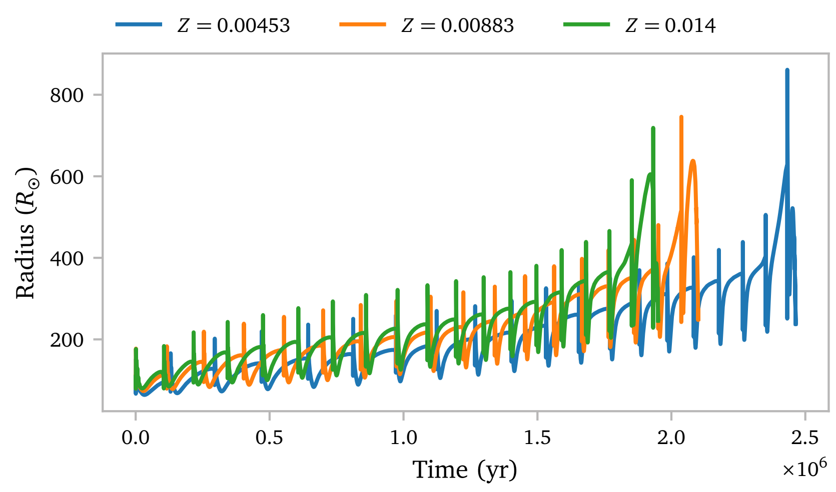
Notes Week 21

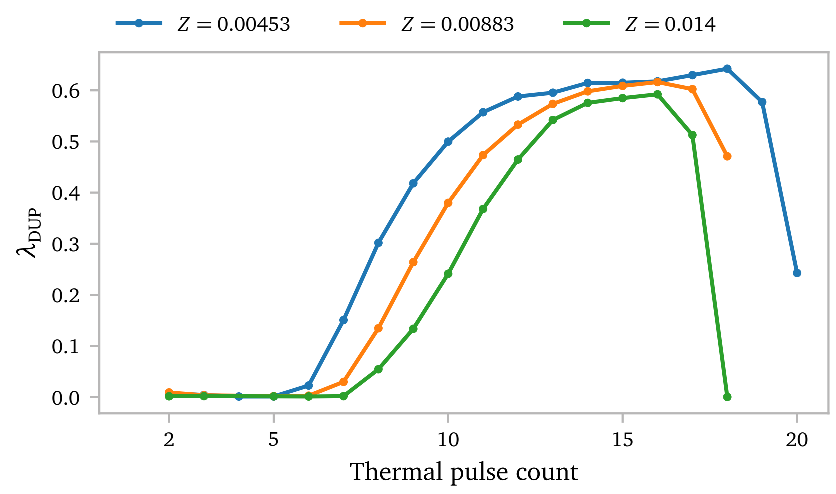
a. Dredge-up and Abundances

To start, I want to plot the radius evolution for the models.



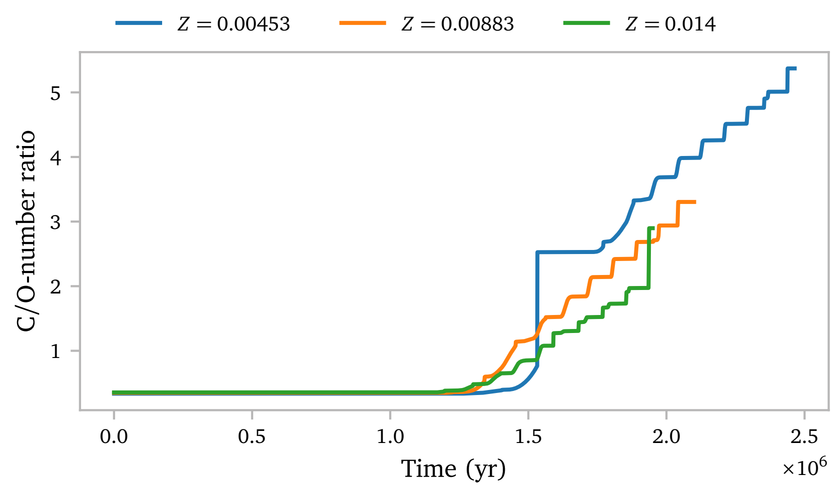
All seem fine, and the general trends that I saw last week seem to hold for the lower metallicity star as well.

Now, more interestingly, let me plot the dredge-up efficiencies.



It appears that dredge is slightly more efficient for lower metallicity stars. This is in line with literature predictions (Herwig, 2005; Karakas et al., 2002). Metallicity also seems to have an inverse relation with the number of thermal pulses that the star will experience.

Below I plot the C/O-number ratio for the different models.



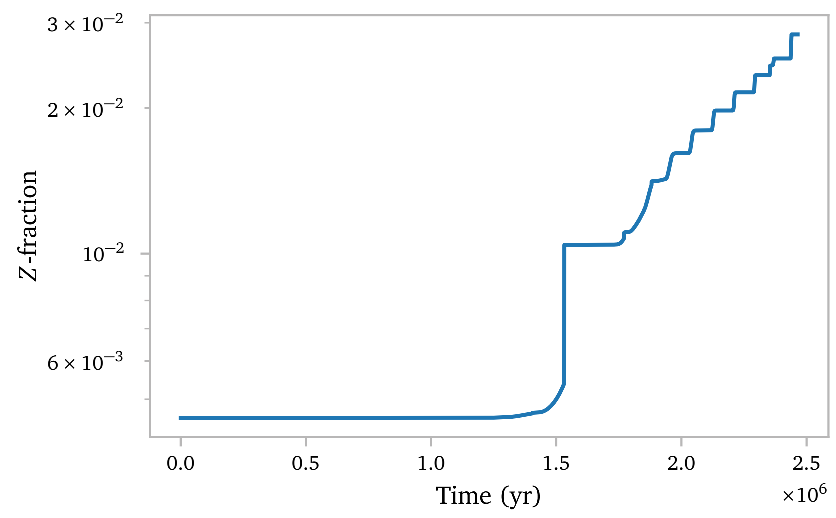
Both the solar model and the $Z = 0.0883$ model look as expected, with the subsolar model achieving a slightly higher C/O-ratio, as expected.

The $Z = 0.00453$ model shows the strong bump that we saw in some of the earlier higher mass models as well.

Since I have a lot of profiles for this star, we can try to work out what is happening[1] .

[1] Unfortunately though, I do not have all of the abundances for all of the profiles, so we need something else that shows the same behaviour.

Let me start by plotting the Z-fraction at the surface of all profiles for the low-metallicity model.

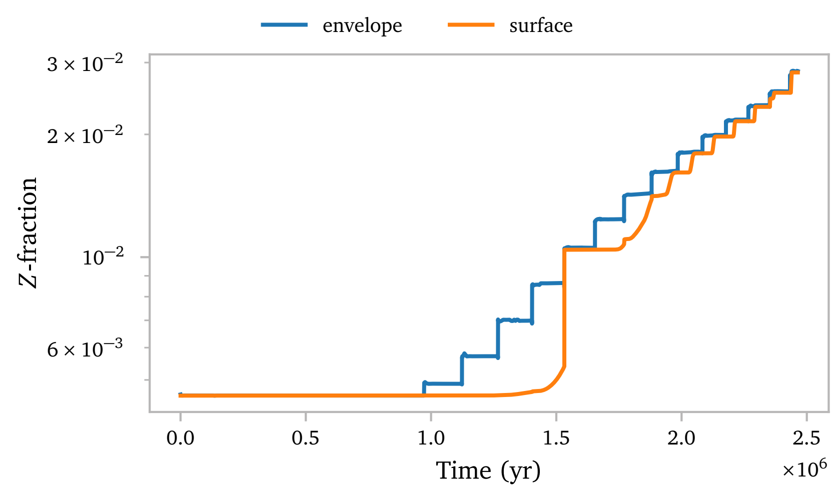


This looks a lot like the plot above, which is good, because that means we can use this Z-fraction as a way to understand the sudden jump.

I really like this interactive figure. It shows:

1. Repeated hydrogen shell burning followed by unstable helium burning.
2. The creation of metals during he-burning followed by dredge up.
3. The loss of the envelope
4. The effect of the superadiabatic sub-surface layers on the metal abundances at the surface.

Clearly the abundance at the surface is not necessarily the abundance in most of the envelope. Below, I try to show that even the model with the sudden jump in surface C/O-ratio still has a well-behaved envelope abundance over time.



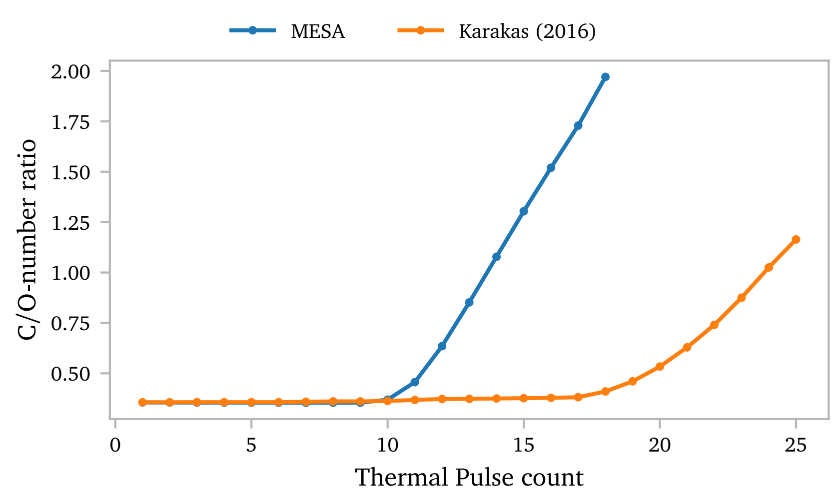
The envelope abundance follows a much more predictable staircase-like pattern.

Since the superadiabatic subsurface layers contain very little mass, the envelope value should provide a much better estimate for the actual abundances that get transferred during RLOF and wind mass transfer.

b. AGB-Nucleosynthesis

I think Karakas & Lugaro (2016) provide abundance data for every thermal pulse for a grid of initial masses and metallicities.

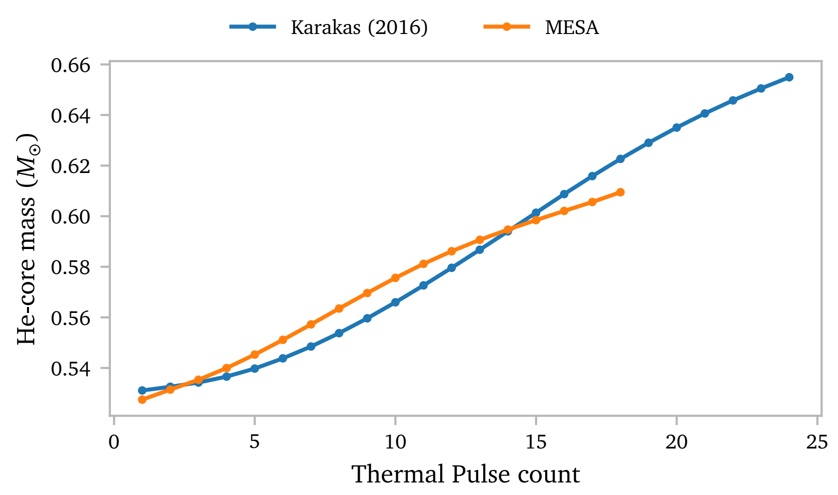
Unfortunately, the metal-poor star they provide has $z = 0.007$, so I cannot trivially match it with a MESA model. For the solar metallicity stars, I *can* do this.



The C/O-ratios do not align *at all*. Not only does the MESA model experience way fewer pulses, it also starts to enrich its envelope much earlier, and the rate at which it enriches its envelope is much faster too.

I think this is quite concerning. It does not seem like we can easily match my MESA model to a Monash model from Karakas & Lugaro (2016). The only thing that is truely consistent is the starting C/O-ratio, but this is expected as both models use the Asplund et al. (2009) solar model.

The core masses are also quite different.

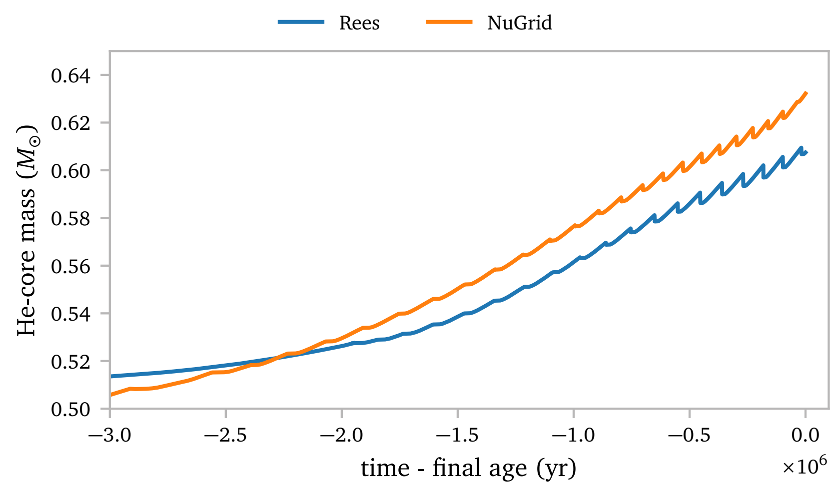


The most notable difference is obviously that the core mass keeps growing for the Karakas & Lugaro (2016) models long after the MESA model has lost its envelope.

The MESA model also seems to grow its core mass quicker than the Karakas & Lugaro (2016) model.

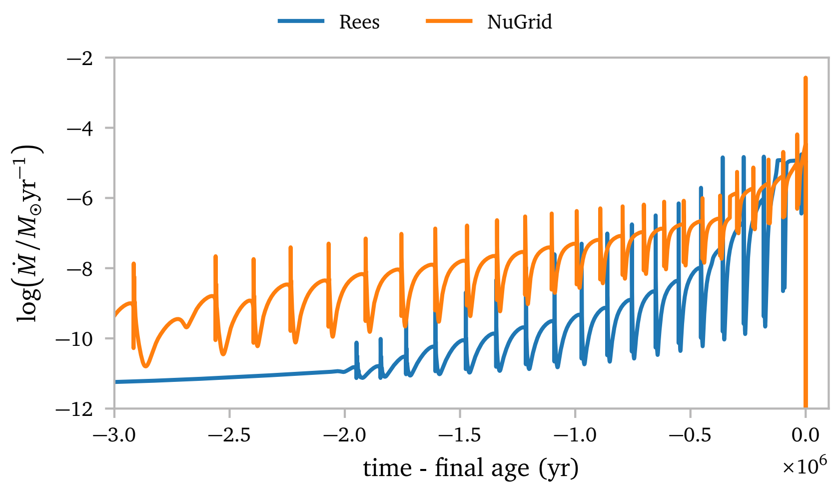
An alternative could be using the NuGrid data.

As a first test, I plot the He-core mass as a function of age



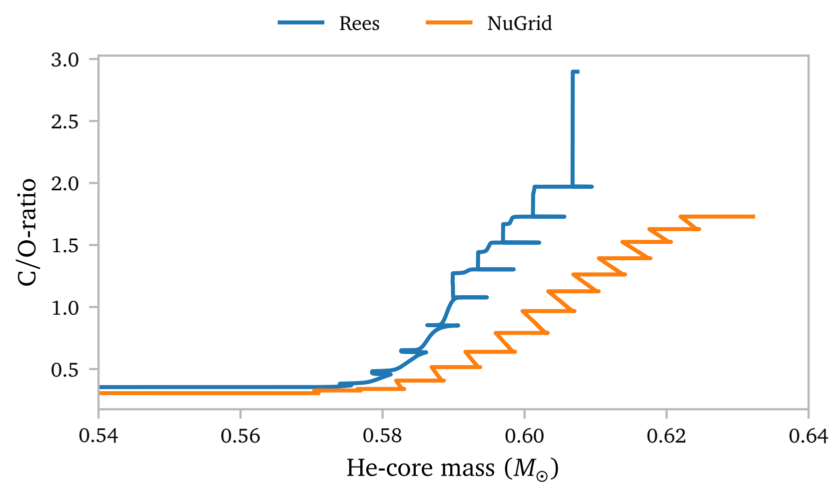
This at least seems a bit closer, but it is still far from identical.

Apart from the core mass, the mass loss rate is very different too.



NuGrid uses Blöcker + Reimers wind, while we use Vassiliadis & Wood (1993). This results in many more thermal pulses in the NuGrid model, simply because the wind is much less strong for much more of the evolution.

Let me compare the C/O-ratio for the Rees models and the NuGrid models.



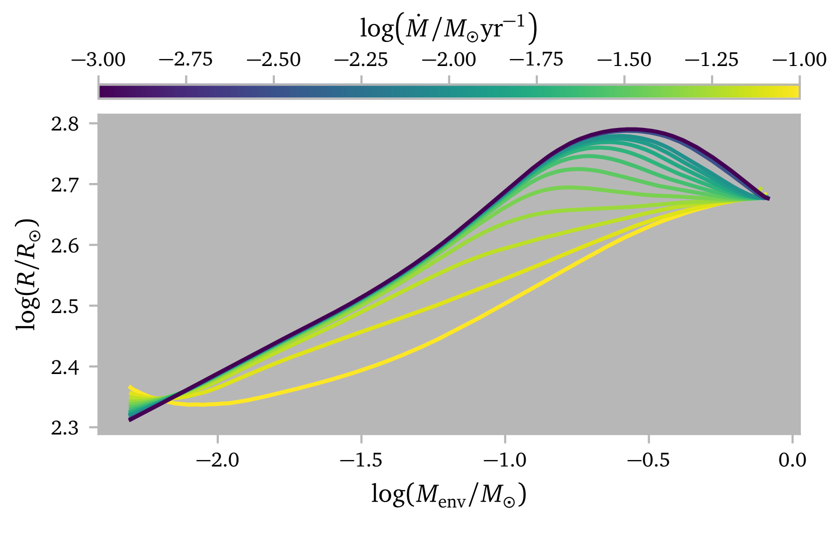
Again, these results are quite different, *although* the NuGrid models do predict larger C/O-ratios than the Monash models.

To me, it seems very difficult to match abundances from a thermal pulse from the Monash models or the NuGrid models to my MESA rees models since they are so different.

c. Constant Mass-loss Rates

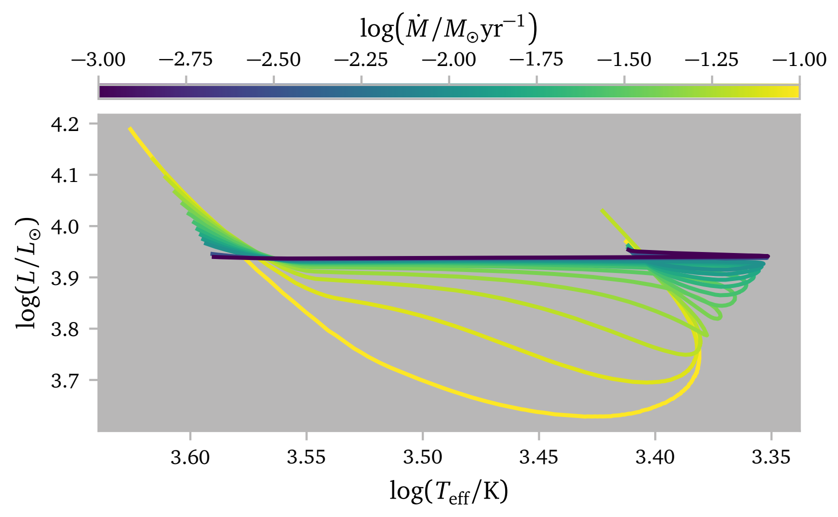
I an attempt to better understand the radial response of the TPAGB star to mass loss, I have simulated a number of models with various *constant* mass loss rates ranging from $10^{-3}\ M_\odot/\mathrm{yr}$ to $10^{-0}\ M_\odot/\mathrm{yr}$. The models losing more than $10^{-1}\ M_\odot/\mathrm{yr}$ however, failed.

Below I plot the results.



We see a similar behaviour as for the detailed binary simulations, where the models with the most extreme mass ratio, and thus the strongest mass loss rate would dip before reaching the post-AGB sequence.

There are minor differences though, in particular, the models with a constant mass loss rate do not seem to decrease their radius quite as much as the models with variable mass loss rate.



The HR diagram shows the move to lower luminosities clearly.

There is also very clearly something weird going on towards the end of the evolution, where the stars start to move to higher luminosities. I suspect this might have something to do with the large mass-loss rates being sustained when the envelope becomes *very* thin.

For these models, I saved profiles logarithmically spaced in envelope mass. This allows for some more detailed inspection of the stellar interior although I am not sure if this was really very helpful. Below I tried to summarize my findings.

The temperature gradient does not change significantly with mass loss rate.

The same is true for the thermal time to the surface.

The same is true for the mixing and nuclear burning.

The same is true for the pressure as well as the individual components of the pressure.

The same is true for the temperature.

The opacity is larger in the envelope of models that lose mass more quickly.

The density is larger in the envelope of models that lose mass more quickly. The entropy stays a bit more flat.

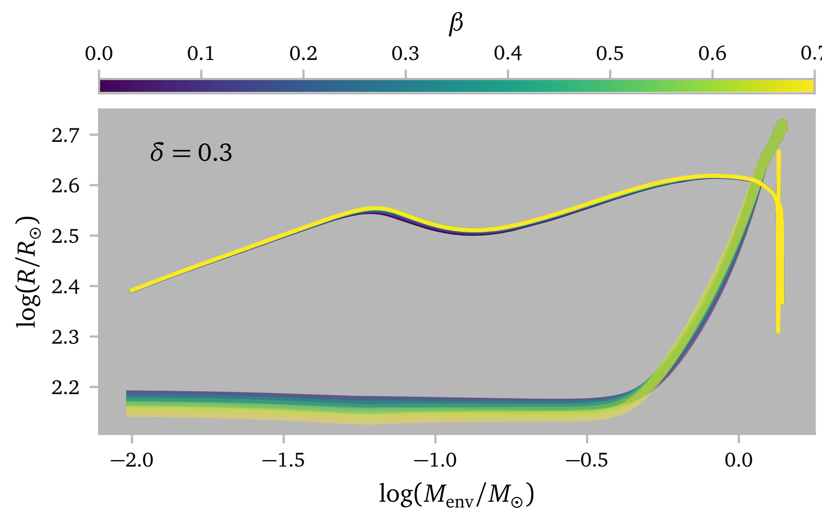
But most significantly, the gravitational luminosity changes a lot for the different models.

d. Conservation Grid

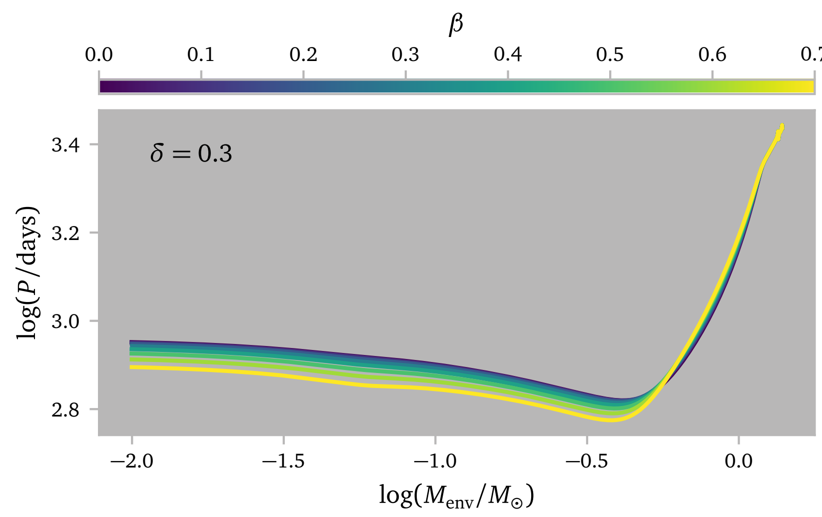
d.1 Varying β

I ran some simulation of a single binary model with different assumptions regarding the conservation of mass transfer. All models start with an initial Roche lobe radius of about $R_{\mathrm{RL}} \approx 550\ R_{\odot}$ and $q = 0.5$. I used $\gamma = 1.44$ for the models to simulate L_2-outflows.

Let me start by showing the effect of β for models with a constant value of $\delta = 0.3$.



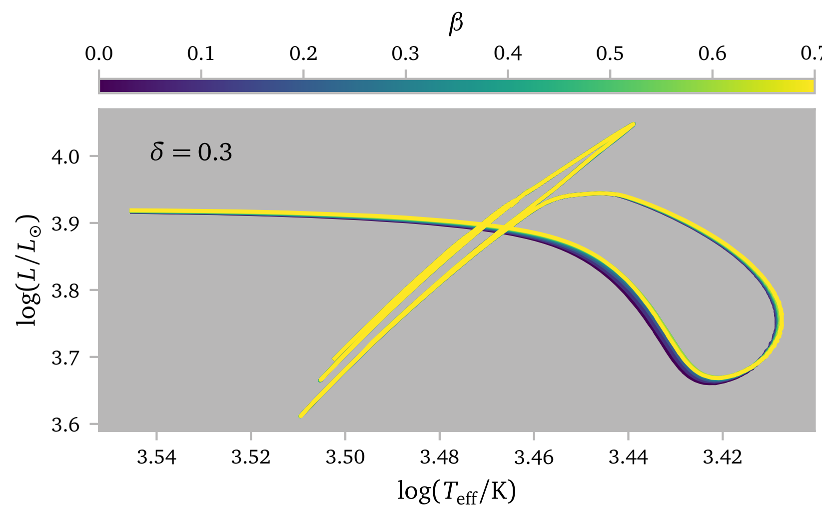
The effect on the star radius is minimal, while a larger β tends to result in somewhat smaller final Roche lobe radii. This corresponds to shorter periods too, which I show in the figure below.



This shows that the periods are shorter for models with larger β. This is also what I expect from the analytical results from Soberman et al. (1997).

Since the radius of the star barely seems to change for different values of β, I do not expect the post-AGB approach to differ.

In the figure below I plot the HR diagram for these models.

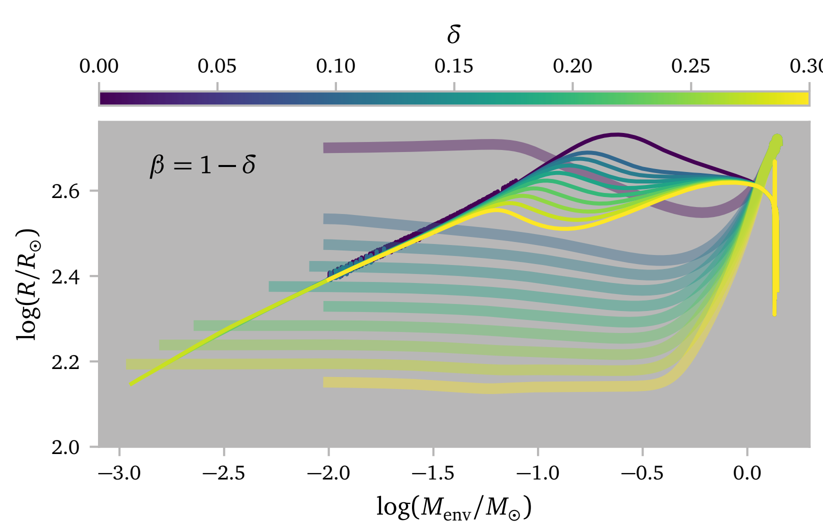


Clearly, there is very little difference in how the star behaves.

d.2 Varying δ

Perhaps more interesting is to see the effects of δ. In this case, I change δ and set β such that $\delta + \beta = 1$[2].

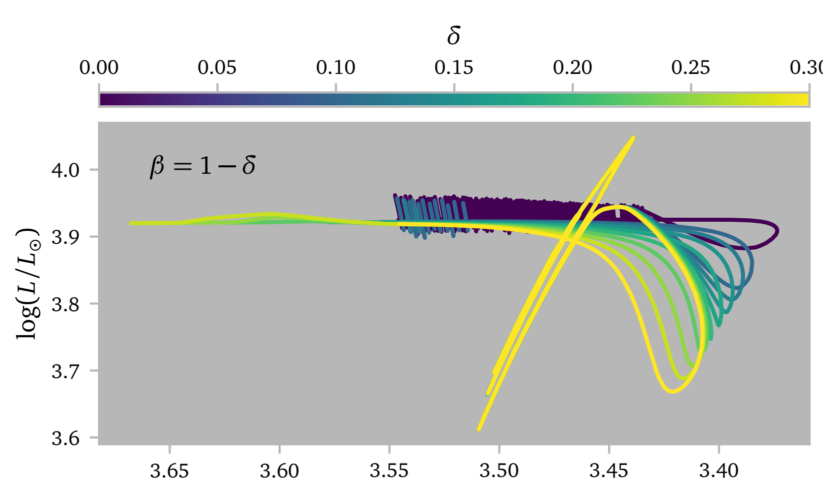
2 So no accretion.



Clearly, changing δ has a much larger effect than β.

Some models stop because the binary spirals in. I have not plotted these.

Some models only ran until $M_{\mathrm{env}} = 0.01 M_{\odot}$ because I forgot to remove the MESA default envelope mass stopping condition. The models are removing clearly behave as expected, stopping only once RLOF has well and truely stopped.



We can see the clear shift to the new Hayashi line. Another interesting thing is that the loops seem to move in opposing directions. for different δ.

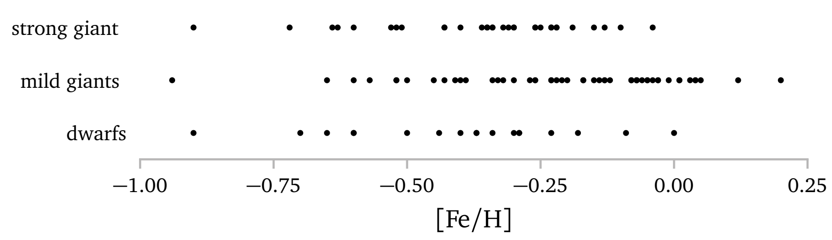
This is more easily visible in the following interactive figure where I only plot four of the many models[3].

3 Because the other models accidentally overwrite the first profiles.

e. Barium Giants vs Dwarfs

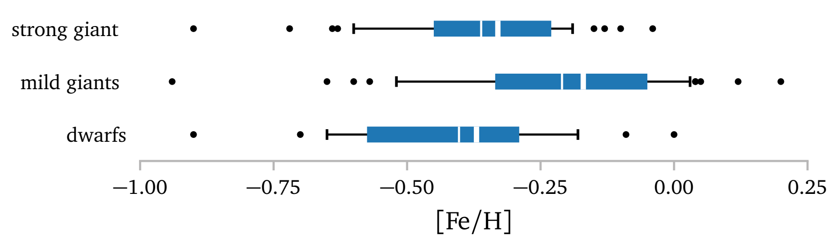
I split the dataset from Escorza & De Rosa (2023) and Jorissen et al. (2019) into three groups: dwarfs, mild giants and strong giants.

Below I show the metallicities of all of the groups.



Clearly, all groups have quite a large spread, going from nearly $z = z_{\odot}/10$ to almost $z = 2z_{\odot}$. The mild giants, on average appear to have a slightly larger metallicity, while the strong giants and dwarfs look similar.

Below I plot a boxplot in an attempt to better show the differences between the distributions.



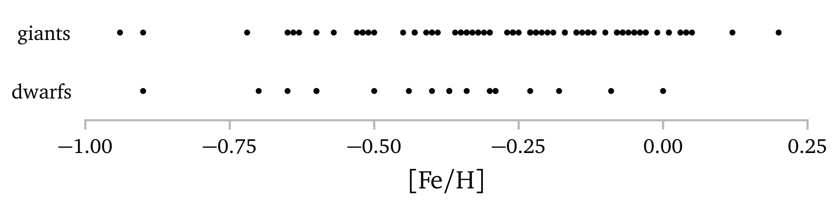
The whiskers indicate the 10% and 90% regions. The box shows the 25% to 75% region. The thick line shows the median and the thin line shows the mean. Dwarfs are slightly metal poorer than giants. The mild giants are more metal rich than the strong giants. This makes sense.

The tabulated metallicity values are

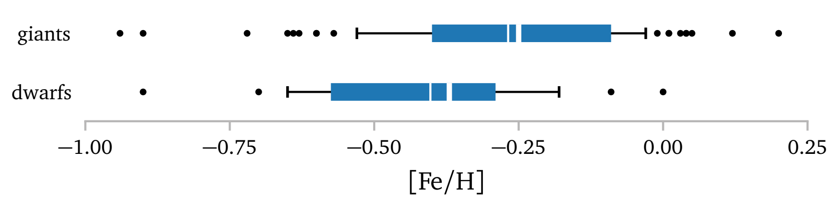
Type	10%	25%	mean	median	75%	90%
strong giants	−0.63	−0.45	−0.36	−0.33	−0.23	−0.15
mild giants	−0.52	−0.335	−0.21	−0.17	−0.05	0.03
dwarfs	−0.67	−0.575	−0.40	−0.37	−0.29	−0.15

Thus far, I have simulated stars with $[\mathrm{Fe/H}] = 0$, -0.2, -0.28 and -0.49. These values do not align particularly well with the subgroups, but they were based on the complete sample. I am not sure which value would be best to choose.

Now, I want to group the giants.



Clearly, there are many more giants than dwarfs. This makes is difficult to directly compare the distributions, but from the plots above, I expect slightly more metal-rich giants.

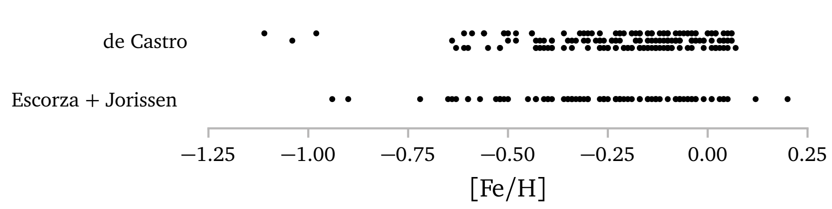


As expected, the giants have larger metallicities.

Type	10%	25%	mean	median	75%	90%
giants	−0.56	−0.40	−0.27	−0.25	−0.09	−0.01
dwarfs	−0.67	−0.575	−0.40	−0.37	−0.29	−0.15

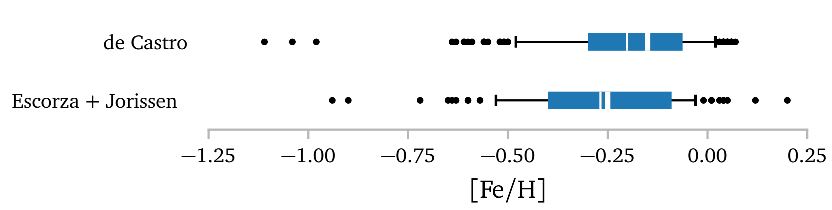
We should now be able to compare the giants to the results from de Castro et al. (2016). They looked specifically at giants and their mean is $[\mathrm{Fe/H}] = -0.20$, so slightly more metal-rich.

Lastly, let me compare the giant sample from de Castro et al. (2016) with the sample from Escorza & De Rosa (2023) and Jorissen et al. (2019). Let



Clearly, the sample from de Castro et al. (2016) has many more stars, but it is not immediately clear if the distributions are very different.

me plot some boxplots again.



Maybe the sample from de Castro et al. (2016) has slightly more metal-rich stars, but the overall, the distributions look similar.

References

Asplund, M., Grevesse, N., Sauval, A. J., & Scott, P. (2009) *The Chemical Composition of the Sun*. ARA&A47, 481.

de Castro, D. B., Pereira, C. B., Roig, F., et al. (2016) *Chemical abundances and kinematics of barium stars*. MNRAS459, 4299.

Escorza, A. & De Rosa, R. J. (2023) *Barium and related stars, and their white-dwarf companions. III. The masses of the white dwarfs*. A&A671, A97.

Herwig, F. (2005) *Evolution of Asymptotic Giant Branch Stars*. ARA&A43, 435.

Jorissen, A., Boffin, H. M. J., Karinkuzhi, D., et al. (2019) *Barium and related stars, and their white-dwarf companions. I. Giant stars*. A&A626, A127.

Karakas, A. I., Lattanzio, J. C., & Pols, O. R. (2002) *Parameterising the Third Dredge-up in Asymptotic Giant Branch Stars*. PASA19, 515.

Karakas, A. I. & Lugaro, M. (2016) *Stellar Yields from Metal-rich Asymptotic Giant Branch Models*. ApJ825, 26.

Soberman, G. E., Phinney, E. S., & van den Heuvel, E. P. J. (1997) *Stability criteria for mass transfer in binary stellar evolution.*. A&A327, 620.

Vassiliadis, E. & Wood, P. R. (1993) *Evolution of Low- and Intermediate-Mass Stars to the End of the Asymptotic Giant Branch with Mass Loss*. ApJ413, 641.

A Additional Figures